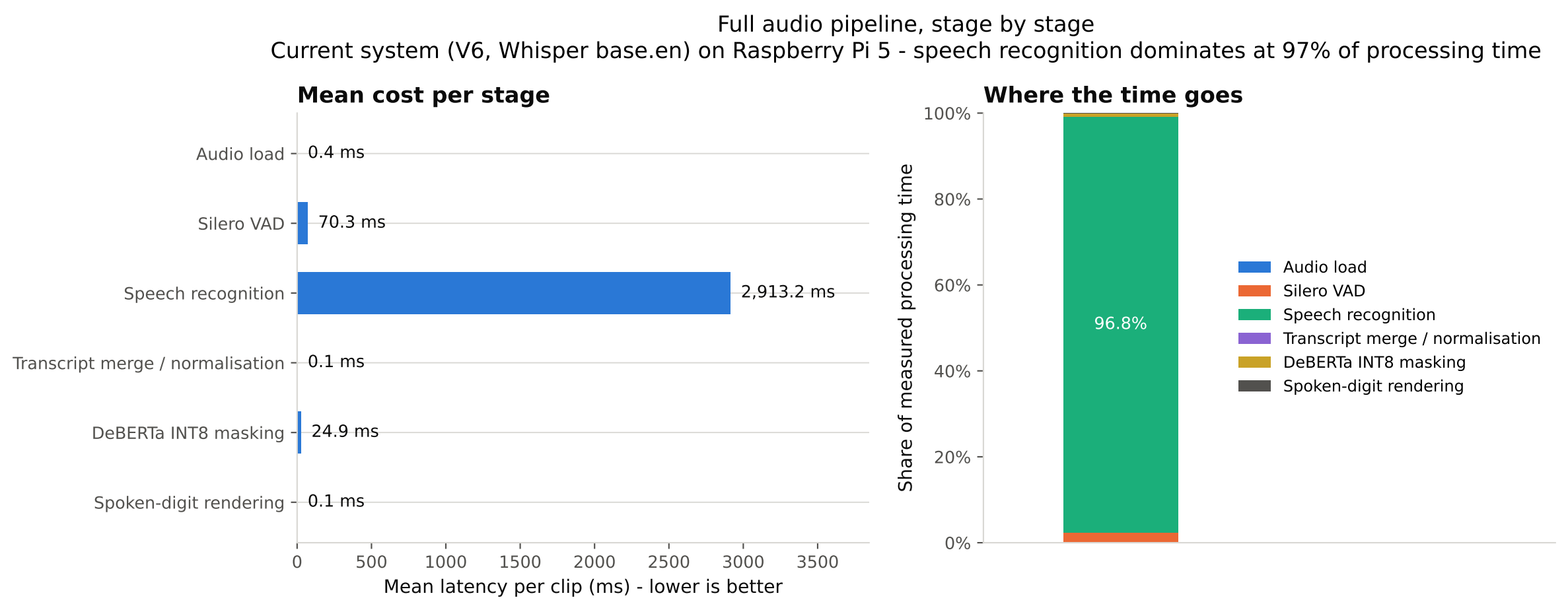
Full audio pipeline, stage by stage
Current system (V6, Whisper base.en) on Raspberry Pi 5 - speech recognition dominates at 97% of processing time
Mean cost per stage
Audio load
0.4 ms
Silero VAD
70.3 ms
Speech recognition
2,913.2 ms
Transcript merge / normalisation
0.1 ms
DeBERTa INT8 masking
24.9 ms
Spoken-digit rendering
0.1 ms
0
500
1000
1500
2000
2500
3000
3500
Mean latency per clip (ms) - lower is better
Where the time goes
Share of measured processing time
100%
80%
60%
40%
20%
0%
96.8%
Audio load
Silero VAD
Speech recognition
Transcript merge / normalisation
DeBERTa INT8 masking
Spoken-digit rendering